CAFÉ MANAGEMENT SYSTEM

Mini Project Report

Submitted by	Kirthana.K
USN	1RUA25BCA0050
Project type	Java Desktop Application
Tools and technologies	Java Swing, JDBC, MySQL, Apache NetBeans, Maven

PROJECT OVERVIEW

The Café Management System is a Java-based desktop application designed to maintain a café's menu inventory. The graphical user interface is built using Java Swing in Apache NetBeans. JDBC connects the application to a MySQL database, allowing menu item information to be stored and retrieved.

The application manages five main data fields: item ID, item name, category, price, and quantity. The item ID is generated automatically by MySQL, while the user enters the name, category, price, and available quantity.

1. Objectives

• Create a simple graphical interface for café menu management.
• Connect a Java application to MySQL using JDBC.
• Insert new menu items and display saved records.
• Update item details and delete records when required.
• Verify that changes are stored in the database.

2. Software and Technologies

Java: application logic and event handling.
Java Swing: form fields, buttons, and JTable display.
JDBC: communication between Java and MySQL.
MySQL: persistent storage for menu records.
Apache NetBeans / Maven: coding, running, and managing the project and MySQL Connector/J dependency.

3. Database Design

The database used is **cafe_db**, and the table is **menu_items**. The table contains item_id (integer primary key with auto-increment), item_name (text), category (text), price (decimal), and quantity (integer). The screenshots show sample records including Cappuccino, Cold Coffee, Veg Sandwich, Chocolate Cake, and crossant.

4. Application Features

Add Item: saves a new menu item using the values entered in the form.
View Items: retrieves menu records from MySQL and displays them in the table.
Update Item: changes the selected record. The screenshot demonstrates the Cappuccino price changing from 120.00 to 170.00.
Delete Item: removes the selected record from the table and database.
Clear: clears the form fields for the next operation.

5. Implementation and Testing

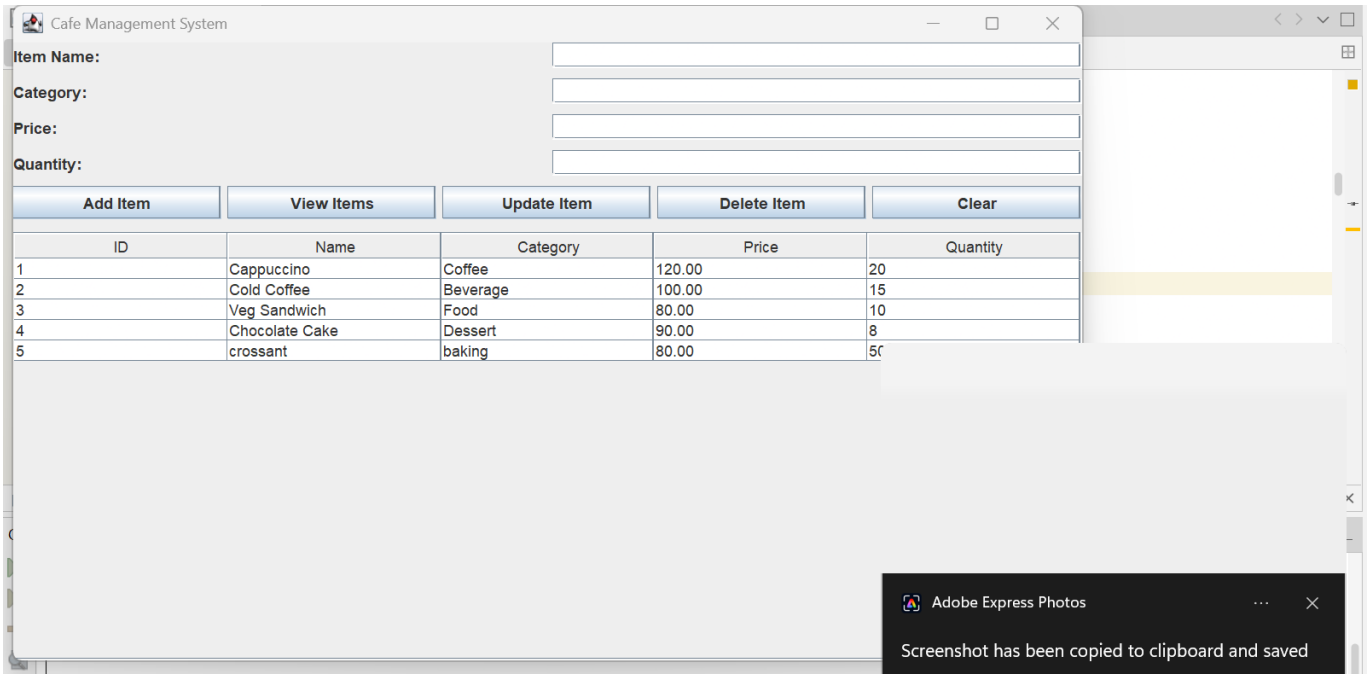
The project is opened in Apache NetBeans as a Maven Java application. The MySQL Connector/J dependency is included in the project, and a JDBC connection is used to access the cafe_db database. The table is loaded using a SELECT query, and insert, update, and delete operations are handled with SQL statements. Prepared statements help pass the form values safely to the database.

The screenshots provide evidence of the running interface, project source code, MySQL table, and SQL result grid. The displayed menu records confirm that data can be retrieved from MySQL. The changed Cappuccino price demonstrates the update workflow.

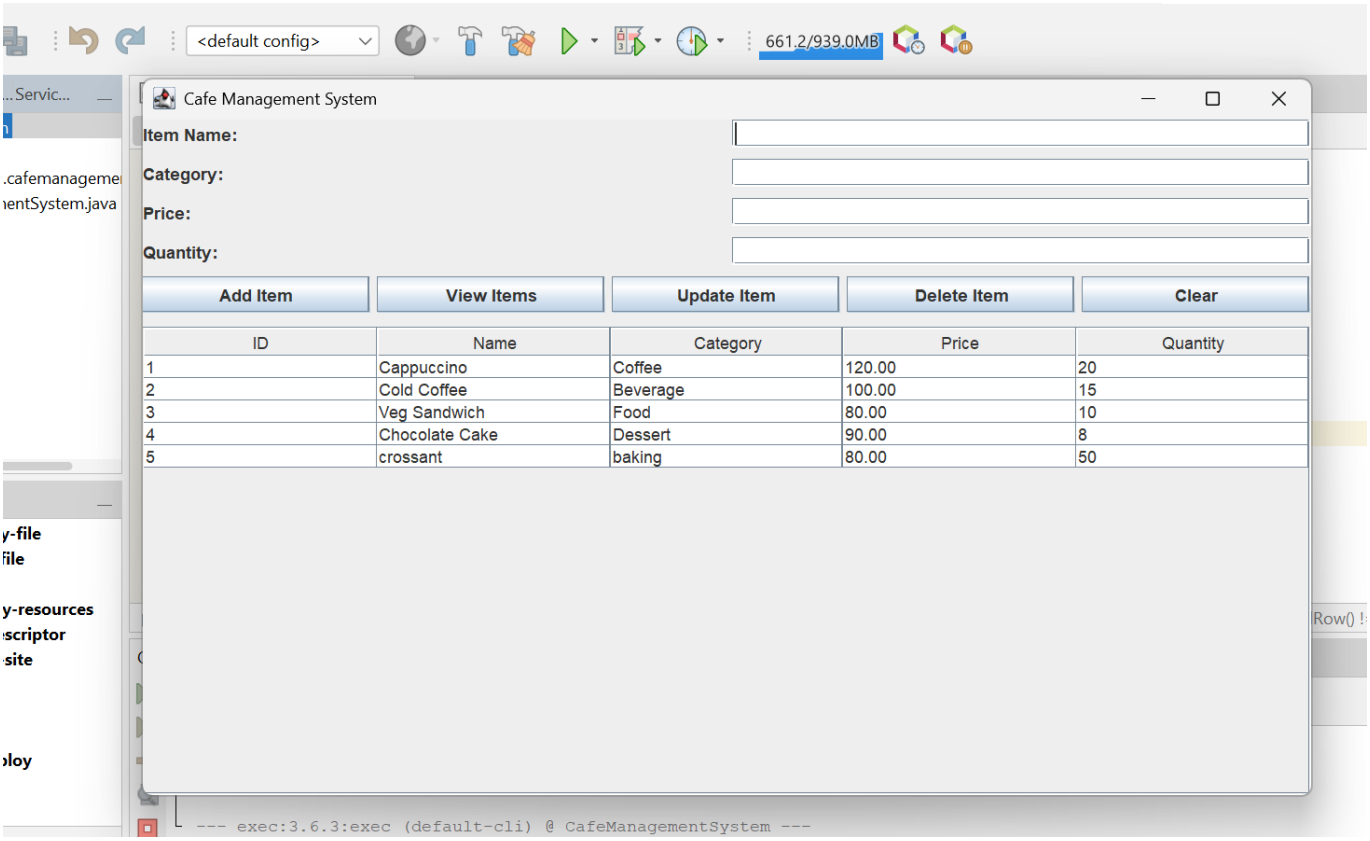
6. Screenshot Evidence

The following pages contain the screenshots supplied for this project. Each caption describes what the screenshot demonstrates.

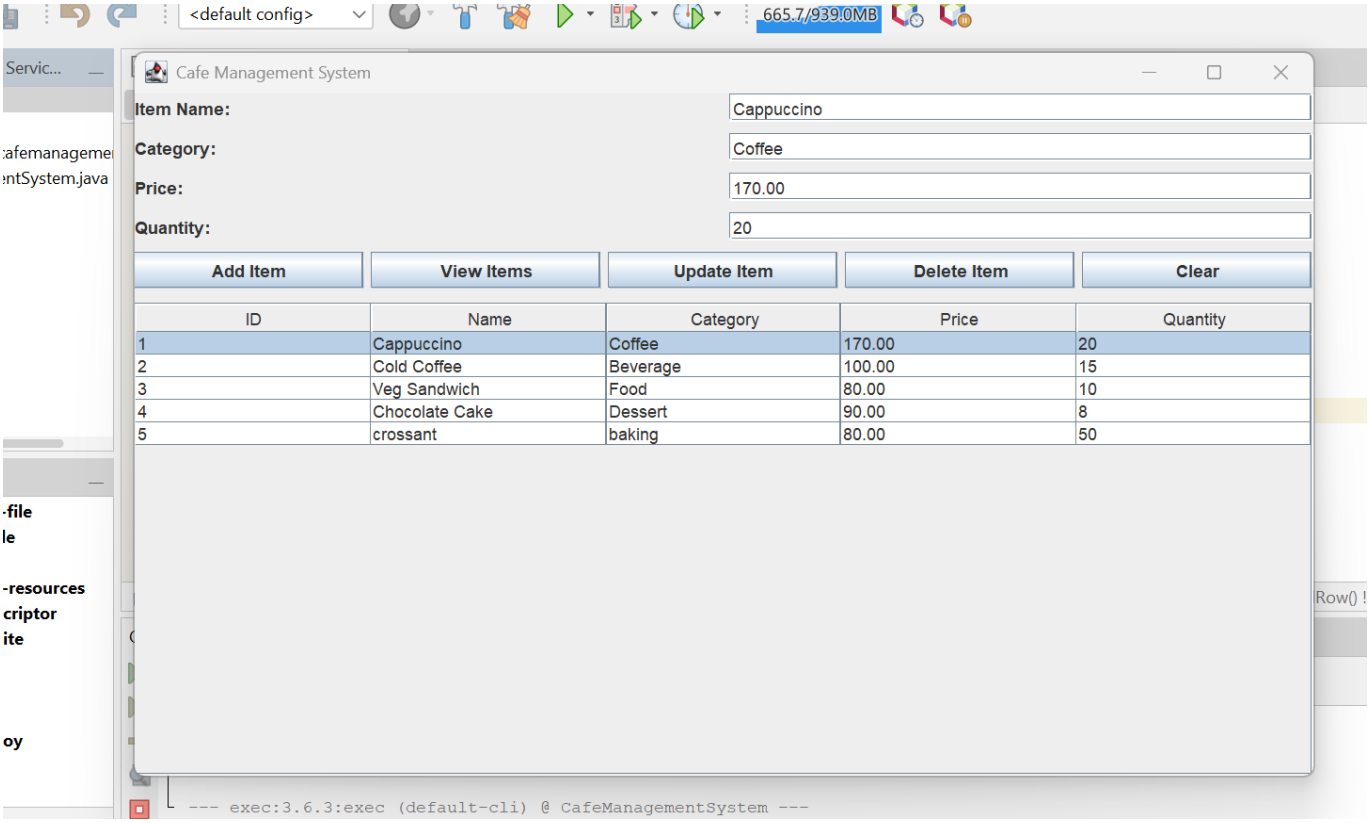
Screenshot 1: Main application window showing the item fields, operation buttons, and menu table.



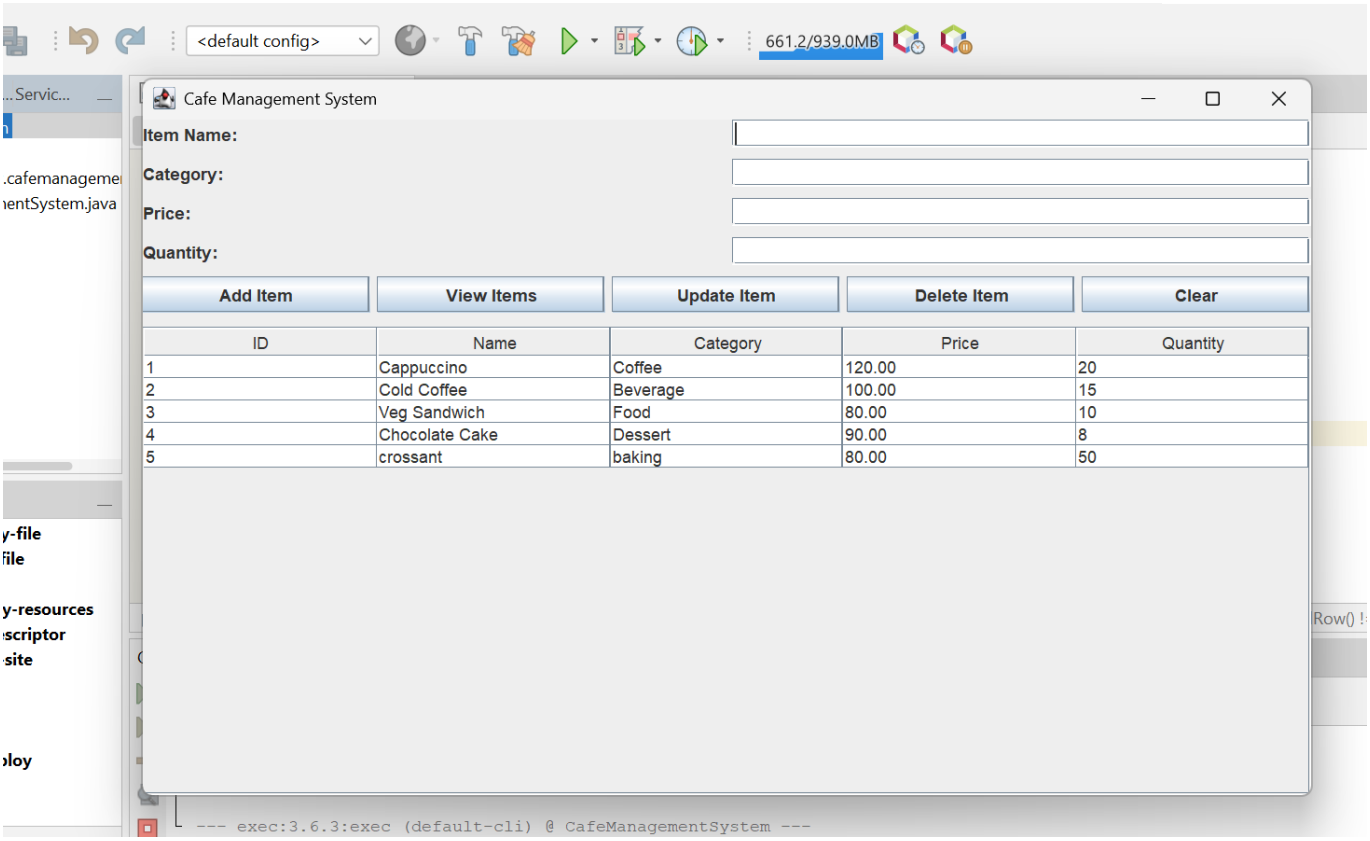
Screenshot 2: Menu item selected in the table and the form populated for editing.



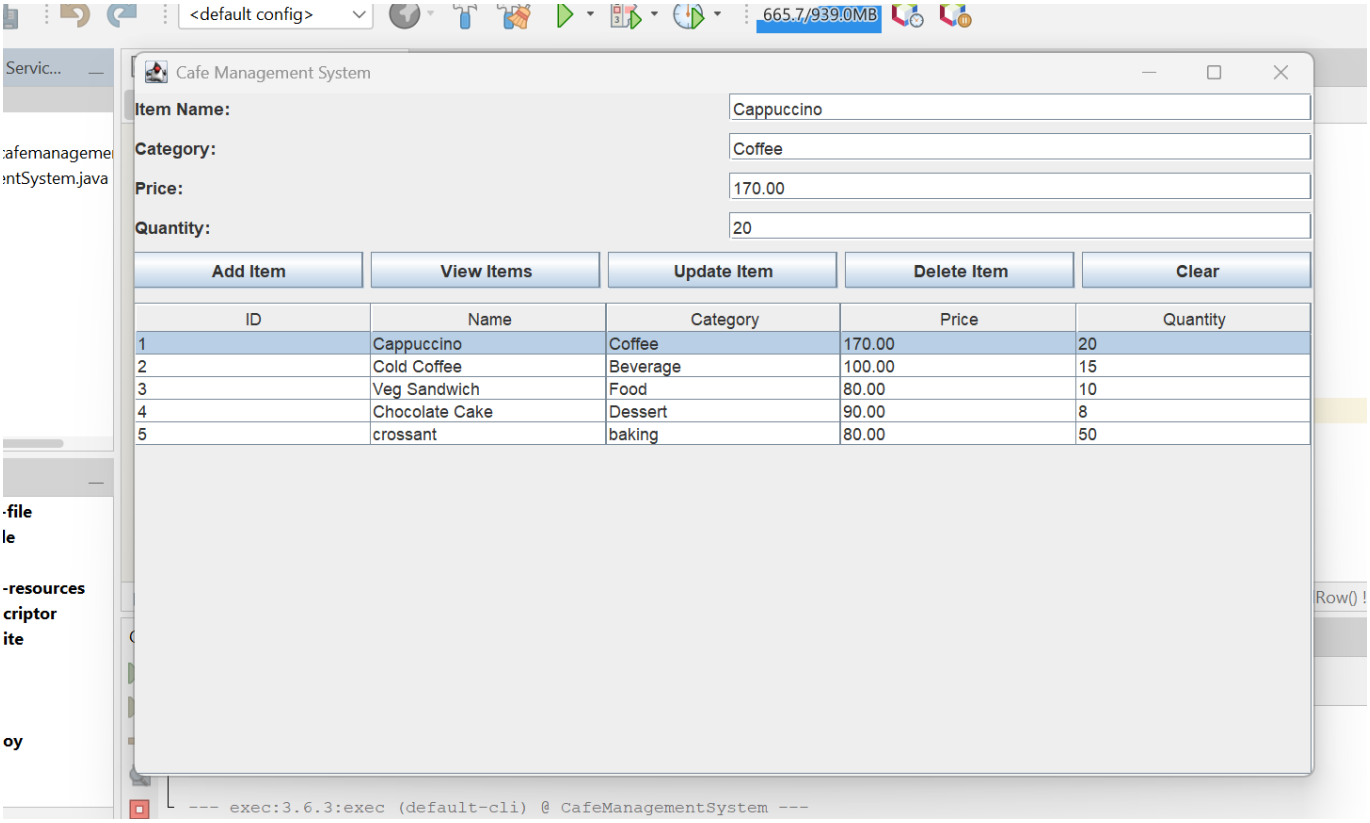
Screenshot 3: Application window showing the menu records after an operation.



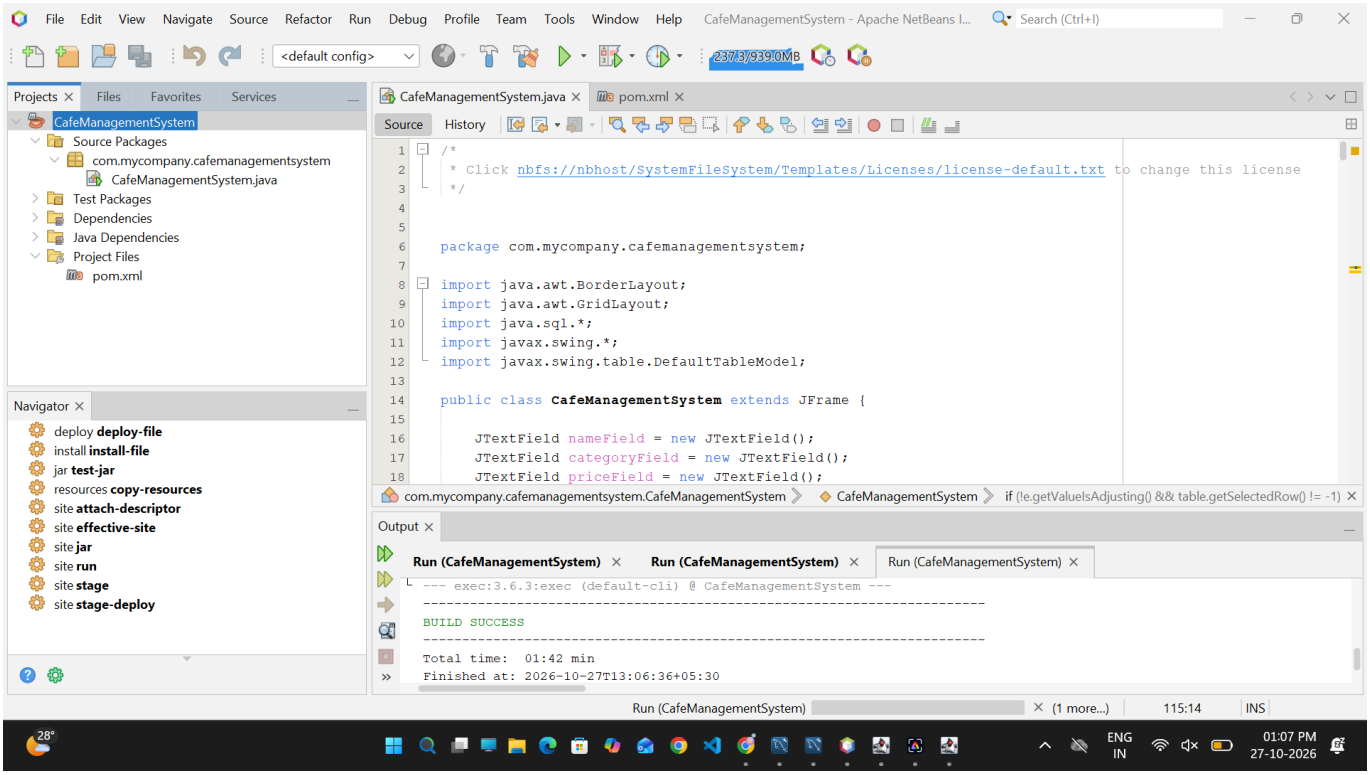
Screenshot 4: Main interface with item names, categories, prices, and quantities.



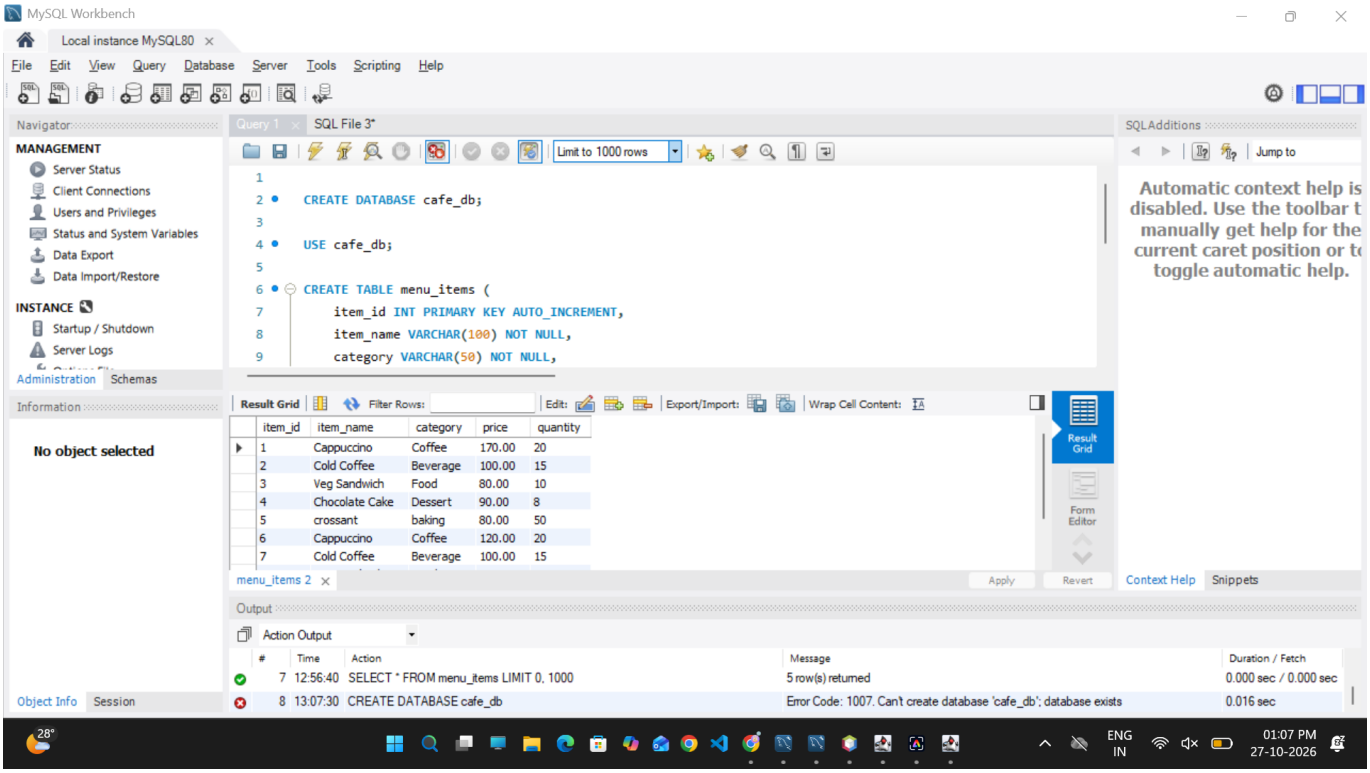
Screenshot 5: Updated Cappuccino record showing the changed price in the table.



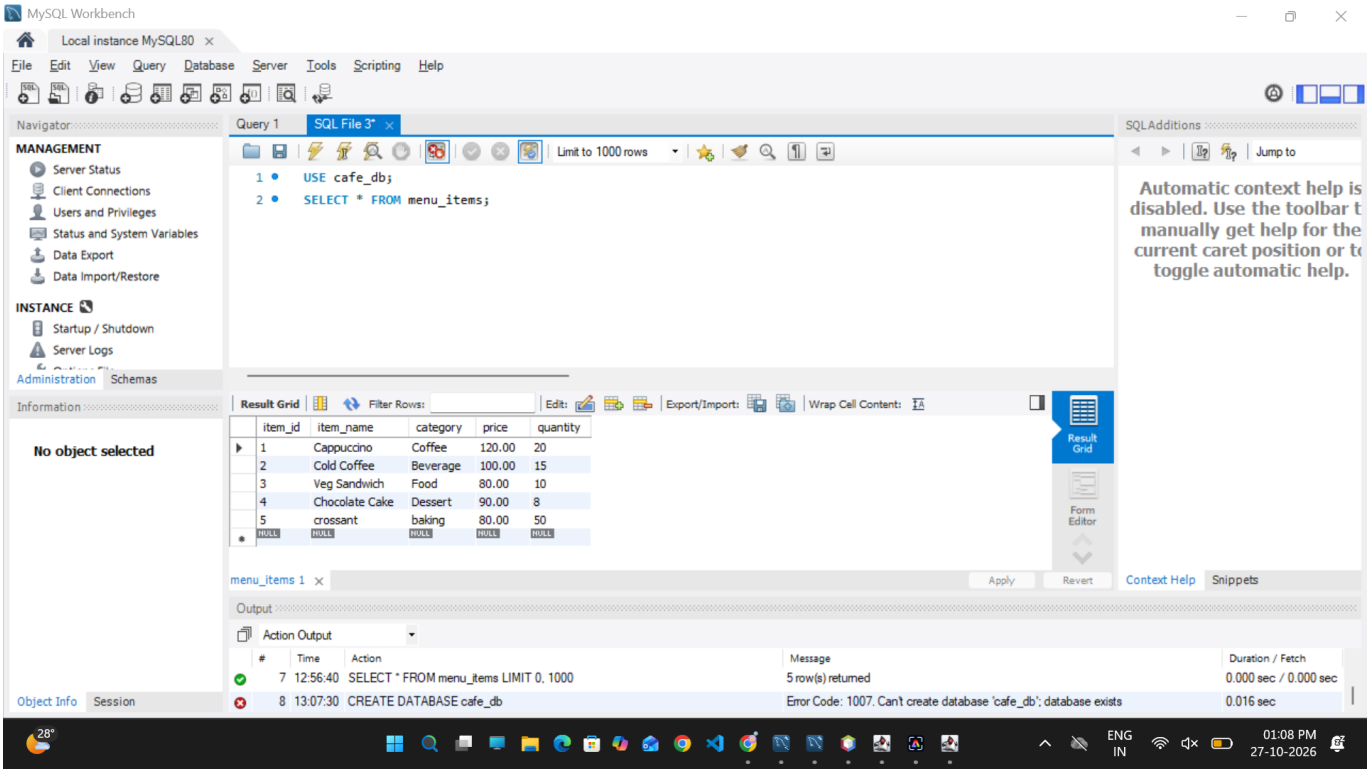
Screenshot 6: Java source code and Maven project structure in Apache NetBeans.



Screenshot 7: MySQL Workbench showing the cafe_db database and menu_items records.



Screenshot 8: SELECT query and result grid showing records stored in MySQL.



7. Conclusion

The Café Management System demonstrates how a Java Swing interface can work with a MySQL database through JDBC. It provides basic create, read, update, and delete operations for café menu items. The project is a practical example of connecting a desktop application to a relational database and displaying stored information in a table.

Submitted by: **Kirthana.K**
USN: **1RUA25BCA0050**